A Synthetic Data-Assisted Satellite Terrestrial Integrated Network Intrusion Detection Framework

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***Abstract*—The Satellite-Terrestrial Integrated Network (STIN) is an emerging paradigm offering seamless network services across geographical boundaries, yet it faces significant security challenges, including limited intrusion prevention capabilities. Federated learning (FL) provides a viable solution by aggregating traffic data from STIN clients (e.g., ground stations and edge routers) to train models for network intrusion detection systems (NIDS). However, satellite and terrestrial domain data's non-independent and identically distributed (non-IID) nature hinders training efficiency and performance. This paper proposes STINIDF, a novel STIN intrusion detection framework leveraging FL-based data augmentation. STINIDF utilizes FL to collaboratively train a conditional diffusion model across STIN nodes while preserving privacy via differential privacy mechanisms, generating global traffic data representative of the STIN distribution. Each node then integrates global and local traffic data to train a local model for NIDS, addressing non-IID challenges by balancing data distribution through data augmentation. Using a simulation environment developed with OMNeT++ and INET, a Satellite-Terrestrial Integrated (STI) traffic dataset was created, including intrusion scenarios such as signal disruption, UDP flooding, and jamming attacks. Experimental results indicate that STINIDF outperforms existing data augmentation-based approaches under non-IID conditions, achieving 96.63%(2.41% ↑) accuracy, 96.71%(3.14% ↑) precision, 96.54%(1.65% ↑) recall and 96.66%(2.7% ↑) F1 score. Furthermore, when compared to methods integrating data augmentation with differential privacy, STINIDF demonstrates an effective balance between privacy preservation and intrusion detection performance, attaining an accuracy of 96.14%(2.57% ↑) and a FID of 17.88(7.41 ↓).**

***Index Terms*—Satellite-terrestrial integrated network, intrusion detection system, diffusion model, federated learning, differential privacy.**

I. Introduction

THE satellite-terrestrial integrated network (STIN) is envisioned as a cornerstone of sixth-generation (6G) networks, enabling heterogeneous services and global network access by supporting extensive traffic transmission for both space access and terrestrial communication [1]. However, expanding satellite networks and rising traffic volumes present critical challenges, including security vulnerabilities, cyber threats, and privacy breaches [2]. To address these issues, network intrusion detection systems (NIDS) have been proposed to monitor and detect malicious activities within satellite and terrestrial traffic flows [3]. To extract features from traffic data and adapt to dynamic environments [4], modern NIDS increasingly leverage artificial intelligence (AI) techniques, such as machine learning (ML) [5] and deep learning (DL) [6]. In STIN, traffic monitors collect traffic data from various infrastructures, such as ground stations (GSs), edge routers, and user terminals. However, it is impractical to train a centralized NIDS because aggregating locally recorded traffic data may leak privacy, and attackers may be able to access sensitive data of GSs in this process [7]. Unlike conventional network environments, STINs serve large-scale commercial and military applications. Therefore, ensuring data privacy is crucial, as any violations could cause significant losses to governments and businesses.

In practice, traffic samples in STIN are collected by the flow monitor distributed across various infrastructures, including GSs, edge routers, and user terminals. Directly training a global NIDS by aggregating these traffic samples is infeasible, as locally recorded data inherently risks privacy breaches. Meanwhile, the intruders may launch adversarial attacks based on personal devices during traffic aggregation to sniff sensitive data in GSs [7]. Compared with other traffic environments like the Internet of Things or industrial control systems, STIN generally serves large-scale commercial or military usage. Therefore, data privacy issues in STIN deserve attention, and any data leakage may cause immeasurable losses to the government and enterprises.

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Federated learning (FL) has emerged as a privacy-aware paradigm, allowing the training of a global AI model across multiple untrusted clients using their local datasets without exchanging data [8]. A representative FL-based algorithm is Federated Averaging (FedAvg), where the parameter server (PS) averagely aggregates client-uploaded model weights [9]. However, training an FL-based NIDS across STIN clients, such as GSs and edge routers, poses two major challenges: data heterogeneity and privacy concerns.

Data heterogeneity, marked by non-independent and identically distributed (non-IID) data distributions across clients, results in inconsistent local models and ineffective aggregation [10]. In STIN, the traffic environment is divided into satellite and terrestrial domains. The satellite domain encompasses traffic exchanged between satellite constellations and GSs, while the terrestrial domain includes flows between GSs, edge routers, and user terminals. GSs record traffic from both domains, whereas edge routers store only terrestrial domain traffic. This imbalance in traffic distribution between GSs and edge routers hinders the effectiveness of FL-based aggregation across STIN clients. To mitigate this issue, data augmentation [11], [12] is applied to synthesize data and balance local training distributions among STIN clients. However, the existing literature fails to consider this approach in the context of STIN-based intrusion detection.

Meanwhile, sensitive information in local traffic can be exposed if intruders intentionally analyze exchanged model parameters through data inversion [13]. To mitigate this risk, differential privacy (DP) [14] is employed to obscure parameters and prevent data inversion. However, few studies have explored the application of DP in real-world scenarios such as STIN.

To tackle these challenges, we propose STINIDF, a novel STIN intrusion detection framework with privacy protection that employs a DP-based conditional diffusion model (DM) [15] for data augmentation. Our contributions include:

- A novel dataset, termed Satellite-Terrestrial Integrated (STI), representing STIN traffic distribution, is collected within a designed simulation scenario using OMNET++ [16] and INET [17].
- A DP-based conditional diffusion model (DP-CDM) is proposed for traffic augmentation with functionality preservation, ensuring that the augmented traffic complies with traffic generation constraints.
- A four-layer FL-based training algorithm, STIN-FedAvg, is designed to train DP-CDM within the STIN environment, incorporating overlapping downloading and multi-level aggregation/local-update properties. Meanwhile, DP is applied during local updates to strengthen privacy protection and mitigate inference attacks.
- STINIDF employs STIN-FedAvg to aggregate DP-CDM, which generates traffic data representing the STIN global traffic distribution. Each STIN node then combines this data with its local data to train a detection model for NIDS, ensuring strong robustness and generalization even in a STIN non-IID scenario.
- Extensive experiments demonstrate that STINIDF outperforms state-of-the-art methods in both traffic generation and detection. Specifically, in the generation aspect, the traffic generated by DP-CDM achieves 17.88 fréchet inception distance (FID), outperforming the second-best method by 7.41. In the detection aspect, NIDS achieves an accuracy of 96.63% in the STIN non-IID scenario, exceeding the second-best method by 2.41%.

The paper is organized as follows. Section II reviews related works in three domains. Section III outlines background and the threat model. Section IV elaborates on the system design of STINIDF. Section VI describes the collected dataset and experiment setups. Section VII presents experimental results and discussions. Section VIII concludes this paper.

II. Related Works

This section discusses related works on STINIDF from multiple perspectives. Since STINIDF is designed to address data heterogeneity challenges, works tackling FL-based non-IID issues are introduced first. As STINIDF incorporates data augmentation within an FL-based system, synthetic data-assisted approaches for privacy preservation are subsequently reviewed. Given that STINIDF is intended for STIN cybersecurity, intrusion detection methods for satellite-related environments are further explored. Finally, a comprehensive comparison across six key issues highlights the distinctiveness of STINIDF within its domain.

A. Non-IID Challenges in Federated Learning

A significant challenge in FL lies in the non-IID data distribution, which disrupts the consistency of the aggregation process across local models [18]. One common approach addresses this by improving the aggregation process. For example, Wang et al. [19] proposed a normalized averaging algorithm to ensure objective consistency while achieving rapid error convergence. Another approach focuses on correcting the local training process by accounting for model deviations. For instance, Mu et al. [20] introduced global prototypes that guide local model training by balancing individual objectives with global optima.

B. Synthetic Data-Assisted Privacy Preservation

Data augmentation is a viable approach to mitigating non-IID challenges by balancing data distribution in FL-based systems [11]. This method shares the model parameters instead of the real data to achieve privacy preservation. For example, FedDPGAN [21] leverages a global generative adversarial network (GAN) [22] trained across distributed nodes to output synthetic data, which are subsequently used to enhance a detection model. However, the training process of GAN often involves frequent data exchanges, leading to significant communication costs. Li et al. [12] proposed a GAN-based framework to generate differentially private synthetic data, which are then labeled by using an iterative pseudo mechanism. While this framework improves consistency and aggregation in their proposed FL-based system, the pseudo-labeling method lacks persuasiveness as it relies on a predefined threshold rather than the intrinsic characteristics

TABLE I
COMPARISON WITH RELATED WORKS IN NETWORK INTRUSION DETECTION

Years	Traffic environments	Detection targets	Strategies	I_1	I_2	I_3	I_4	I_5	I_6
2020 [2]	Satellite-Terrestrial Integrated Networks	Distributed denial-of-service	FL-adapted with complexity optimization	●	●	○	○	◐	●
2022 [23]	LEO satellite networks	Interruption attacks	MLP, CNN, RNN, GRU, LSTM	◐	○	○	○	○	●
2022 [24]	IoT-integrated satellite networks	IoT-related cyber threats	FL- and DL-based framework	◐	●	○	○	◐	○
2022 [28]	IoT Spectrum Sensors	Spectrum sensing data falsification	Federated model	○	●	○	○	◐	●
2022 [29]	IoT systems	Malicious behaviours	FL with clusters of trust	○	●	●	○	●	○
2022 [30]	IoT systems	Minority intrusions	Improved VAE and oversampling	○	○	○	●	○	○
2023 [3]	Satellite-terrestrial integrated networks	Malicious activities	ML/DL models with feature selection	◐	●	○	○	◐	○
2023 [31]	A mesh satellite network	Autonomous-vehicle-related threats	FL-based IDS in a mesh architecture	◐	●	●	○	◐	○
2023 [32]	Satellite-IoT space networks	Malicious data transmission	FL- and ML-based SDN backbone	◐	●	○	○	◐	○
2024 [33]	IoT networks	Suspicious network activities	Collaborative GAN-based FL with DP	○	●	●	●	●	○
Proposed	Satellite-terrestrial integrated networks	Terrestrial/Satellite threats	DM- and FL-based IDS with DP	●	●	●	●	●	●

"○" shows that the method does not pay attention to the issue,
"●" shows that the method fully pays attention to the issue,
"◐" shows that the method partly pays attention to the issue.

of the data. Moreover, these studies predominantly address image classification or medical diagnosis, rather than network intrusion detection, particularly in STINs.

C. Intrusion Detection for Satellite-Related Networks

Li et al. [2] proposed a distributed NIDS on STINs, incorporating an FL-adapted algorithm to optimize recognition rates, packet loss rates, and CPU utilization. However, the non-IID nature of traffic in STINs was not considered. Sitouah et al. [23] explored various deep learning algorithms for detecting attacks in low earth orbit (LEO) satellite networks, focusing on interruption categories such as DoS/DDoS, jamming, and meteorological disturbances. Despite their contributions, the LEO traffic environment is indeed different from STINs, where the satellites are prohibitive to communicate with user terminals or edge routers without GSs. Additionally, DFSat [24] was proposed for intrusion detection on IoT-integrated distributed satellite networks, leveraging an FL-based deep learning framework to achieve higher performance with fewer communication rounds. Nonetheless, the datasets used in their experiments, ToN_IoT [25] and UNSW-NB 15 [26], were generated from terrestrial simulated platforms rather than satellite-related platforms, some key features like signal-to-interference-plus-noise radio (SINR) [27] are not included, and the data distribution is distinct. Furthermore, Azar et al. [3] developed SAT-IDS, a hybrid intrusion detection system for satellite-terrestrial systems, employing AI-based methods with sequential forward feature selection. However, their system is isolated and challenging to deploy in realistic distributed environments like STINs.

D. Work Comparison

Except for the aforementioned review, Table I provides a comparison with other relevant works in network intrusion detection, addressing various issues concerning STINs and related environments. Specifically, I_1 represents the focus on STIN-specific environments; I_2 indicates support for distributed deployment; I_3 reflects consideration of the non-IID challenge; I_4 pertains to the application of data augmentation; I_5 highlights privacy preservation measures; I_6 refers to manual traffic simulation and dataset collection. Our STINIDF aims to comprehensively cover all six issues.

III. PRELIMINARIES

A. Federated Learning

It is a distributed ML-based training strategy with data privacy preservation. One of the most representative FLs is FedAvg [9], where updated local model parameters are averagely aggregated. Concretely, for each communication round $t \in T$, every client $k \in K$ first updates its local model weights w_t^k by its local dataset D_k via stochastic gradient descent (SGD) optimizer [34]. Then, all trained local weights w_{t+1}^k are transferred to PS for global weight aggregation: $w_{t+1} \leftarrow \sum_{k=1}^{K} w_{t+1}^k$. w_{t+1} is also downloaded by each client for its next local training phase. Overall, FL is designed to solve the following risk minimization problem:

$$\min_w \left\{ f(w) \triangleq \frac{1}{K} \sum_{k=1}^{K} F_k(w) \right\} \tag{1}$$

where $F_k(w) \triangleq \mathbb{E}_{x \sim D_k}[F_k(w, x)]$ is the loss function for local update. After T times, the trained global weights contain the knowledge from each D_k without data exchange.

B. Differential Privacy

Considering that the exchanged model weight can still be divulged by some existing analyzing tools, bringing information leakage problems, DP is designed by adding artificial noise to entail concealing sensitive instances within the dataset [13]. Given a randomized mechanism $\mathcal{M}$, it satisfies (ϵ, δ)-DP when any possible output set Ω and any two adjacent databases D, D' hold the following inequality:

$$Pr(\mathcal{M}(D) \in \Omega) \leq e^{\epsilon} Pr(\mathcal{M}(D') \in \Omega) + \delta \tag{2}$$

ϵ is the pre-defined privacy budget to balance the strength of DP. Smaller ϵ means stronger privacy protection and worse data utility, and vice versa. δ represents the risk probability of condition that ϵ can not bound D and D'. In this paper, we utilize the Gaussian mechanism to guarantee (ϵ, δ)-DP for the traffic data, where artificial Gaussian noises $z \sim \mathcal{N}(0, \sigma^2 \mathrm{I})$ are added to the exchanged model weights. The noise scale σ should satisfies $\sigma \geq S\sqrt{2ln(1.25/\delta)}/\epsilon$ where S is the sensitivity of the model function $f_w(\cdot)$ given by $S = max_{D,D'} \|f_w(D) - f_w(D')\|$. For Gaussian-based DP in DL, the differential private stochastic gradient descent (DP-SGD)

[35] with (ϵ, δ)-DP requires a single model update clipped to C (i.e., the gradient norm bound) with noise Cz addition (In this paper, S is equals to C for single sample update), defined as follow:

$$w \leftarrow w - \frac{\eta}{M}\left(\sum_{i=1}^{M}\frac{d\mathcal{L}_w(x_i)}{dw} min\left(1, C/\|\frac{d\mathcal{L}_w(x_i)}{dw}\|_2\right) + Cz\right) \tag{3}$$

where η is the learning rate, M is the sample amount and $\mathcal{L}_w(\cdot)$ is the loss function of the model.

C. Diffusion Model

DM [15] is famous for generating stable features for constructing new instances with high quality such as multi-media materials. Denoted as p_w, DM generally includes forward and backward processes. The forward process gradually converts x_0, sampled from real data distribution $q(x)$, into Gaussian noise vectors $x_N = \mathcal{N}(0, \mathrm{I})$ in each step $n \in [1, 2, \ldots, N]$. This process is also known as posterior probabilities: $q(x_n|x_{n-1})$. Given a variance schedule $\{\beta_n \in (0, 1)\}_{n=0}^{N}$, it is fixed into a Markov chain for $x_0 \rightarrow x_N$:

$$q(x_n|x_{n-1}) = \mathcal{N}(x_n; \sqrt{1-\beta_n}x_{n-1}, \beta_n\mathrm{I}) \tag{4}$$

This formula can be optimized by the Gaussian noise reparameterization technique, where x_N is directly calculated from x_0. As a result, Equation 4 is converted into Equation 5 given $\alpha_n = 1 - \beta_n$ and $\bar{\alpha_N} = \sum_{n=1}^{N}\alpha_n$:

$$q(x_N|x_0) = \mathcal{N}(x_N; \sqrt{\bar{\alpha_N}}x_0, (1-\bar{\alpha_N})\mathrm{I}) \tag{5}$$

The reverse process samples x_N and converts this Gaussian vector back to a new vector x_0 with p_w. This step is known as conditional probabilities: $p_w(x_{n-1}|x_n)$. It is also fixed into a Markov chain for $x_N \rightarrow x_0$:

$$p_w(x_{n-1}|x_n) = \mathcal{N}\left(x_{n-1}; \mu_w(x_n, n), \sum\nolimits_w(x_n, n)\right) \tag{6}$$

where μ_w predicts the forward process posterior mean and $\sum_w(x_n, n)$ equals $\beta_n\mathrm{I}$, which is the variance. μ_w can also be parameterized as:

$$\mu_w = \left(x_n - \frac{\beta_n}{\sqrt{1-\bar{\alpha}_n}}z_w(x_n, n)\right) / \sqrt{\alpha_n} \tag{7}$$

where z_w is an approximator that requires gradient learning for approaching Gaussian vector z given x_n. Finally, the reverse process from x_n to x_{n-1} is specified as:

$$x_{n-1} = \left(x_n - \frac{\beta_n}{\sqrt{1-\bar{\alpha}_n}}z_w(x_n, n)\right) / \sqrt{\alpha_n} + \beta_n z \tag{8}$$

Conclusively, training a DM is to foster reverse approximator z_w to predict n-step z, its empirical loss function $\mathcal{L}_w$ is defined as a denoising score that matches over multiple noise scales indexed by n with the expectation of x:

$$\mathcal{L}_w = \mathbb{E}_{x,n,z}\|z - z_w(\sqrt{\bar{\alpha}_n}x + \sqrt{1-\bar{\alpha}_n}z, n)\|^2 \tag{9}$$

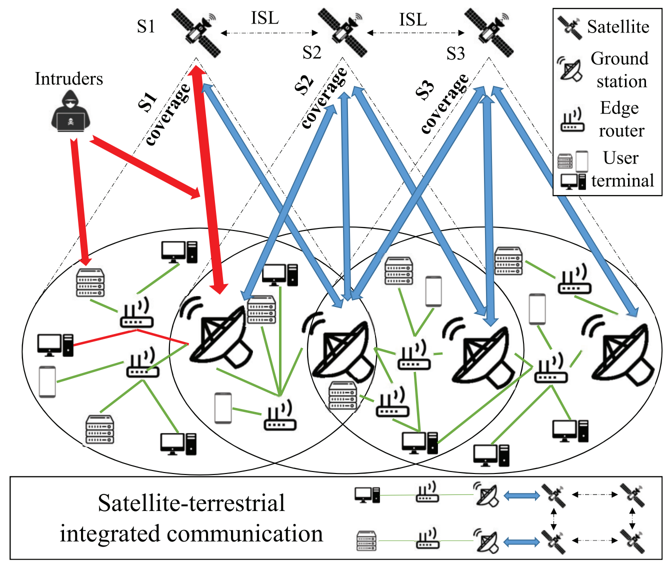


Fig. 1. Overall threat model that includes space, ground, and user segments. The space segment possesses satellite nodes that send messages by ISL with each other; the ground segment symbolizes GS nodes that are responsible for satellite-ground communications, each satellite covers several GSs in a fixed coverage, and the user segment refers to all kinds of terminals that send service requests to GSs. Long-distance communication between two terminals will bypass two GSs and several satellites. Intruders may attack satellites or GSs by infecting the terminals or interfering with satellite-ground communications.

D. Threat Model

The threat model outlined in Figure 1 provides a comprehensive overview of the distributed architecture inherent in a typical STIN, encompassing satellites, GSs, edge routers, and user terminals. The satellites are strategically positioned in intended orbits, facilitating dynamic coverage and communication with GSs, while also establishing inter-satellite links (ISLs) to interact with each other. When two GSs in different areas require a data exchange, these satellites are supposed to work as transponders to transfer information in space. The GSs play a pivotal role in traffic intersections and achieving telecommunication with satellites, ensuring seamless connectivity. They are responsible for summarizing incoming traffic in distinct areas from their connected edge routers. The edge routers in this model are responsible for collecting satellite-based telecommunication from the user terminals. The user terminals refer to various devices like enterprise servers, computers, and telephones. When one of them launches a telecommunication with a terminal in another area, a satellite-terrestrial integrated communication occurs. The communication path will traverse two edge routers, two GSs, and multiple satellites interconnected by ISLs.

Assumptions underpinning the threat model include the prohibition of direct communication between the satellites and non-GS nodes, including the user terminals and the edge routers. This is for improving the leverage of the GSs and mitigating security risks associated with the satellites. Considering that intruders are almost unfeasible to launch an in-space attack that directly interferes with satellites or ISL links in STIN, while every GS is configured with cybersecurity administrators and multiple network protections, the integrity

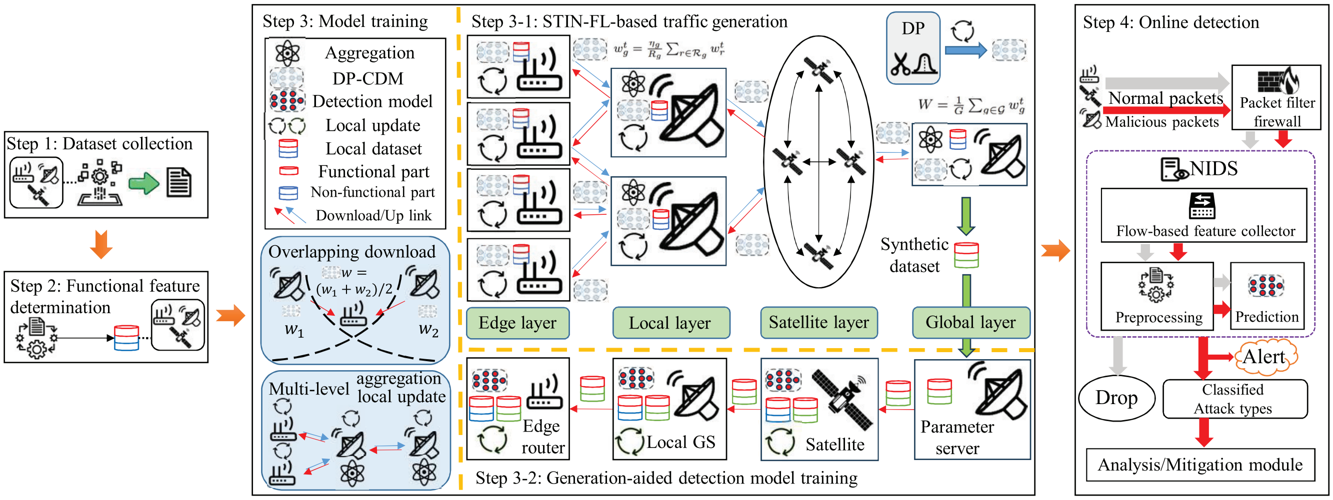


Fig. 2. The system design of STINIDF. After dataset collection and functional feature determination, the model training requires all extracted features to be scattered over a four-layer STIN system. This system aggregates a conditional DP-based diffusion model named DP-CDM based on FL with two properties: overlapping downloading and multi-level aggregation/local update. DP-CDM is supposed to generate non-functional features to form a global dataset together with the functional features in PS. DP is applied in its local update algorithm by gradient clipping and noise adding for higher privacy protection. The edge router, GS, and satellite nodes download and utilize this dataset and the local dataset to train distinctive detection models for online malicious packet detection.

of the satellites, GSs, and ISL-based connections is consequently assumed. However, it is available and effortless for intruders to control any edge routers, user terminals, or infect non-ISL links, as they are trivial and with lower safeguards. Therefore, these nodes and links are deemed untrusted. With the untrusted nature of the edge routers, user terminals, and non-ISL links susceptible to malicious activities, intruders may exploit vulnerabilities and launch cyber attacks on GSs from terrestrial devices or target GS-satellite communications through nefarious means. To safeguard against these cyber attacks within STIN, it is necessary to develop a robust and distributed intrusion detection mechanism. This mechanism is required to achieve both satellite and terrestrial domain traffic detection, together with deployment in each STIN traffic pivot, including satellite, GS, and edge router nodes.

IV. Methodology

The system design of the proposed method is shown in Figure 2. It includes four steps: data collection, functional feature determination, model training, and online detection.

The data collection step generates STI traffic instances with benign and attack labels by a simulated STIN-based platform with OMNET++ and INET. These labeled instances are then utilized for the following experimental validation.

The functional feature determination step meets the requirement in the domain of traffic generation, it determines which part of the features can be transformed by the generative model.

The model training step possesses a STIN-FL-based traffic generation part and a synthetic data-assisted detection model training part. These two parts are implemented under a STIN environment. The first part trains a DP-based conditional DM, named DP-CDM, based on FL across STIN clients with overlapping downloading and multi-level aggregation/local update. DP is applied in every local update process based on gradient clipping and noise adding for higher privacy protection. After weight aggregation into a PS, this DM synthesizes a traffic dataset that reflects the global data distribution of this STIN. In the second part, this synthesized dataset is downloaded by each edge router, local GS and satellite to train its detection model, together with its own local recorded traffic.

The online detection step illustrates the detection process in STIN, where each edge router, local GS and satellite applies its trained detection model as a part of NIDS to monitor the passed traffic with a firewall and a flow collector.

The main symbols in this section are collected in Table II for quick reference.

TABLE II
List of Main Symbols for Proposed Method

	Description		Description
B	Batch size	C	Clip parameter for DP
$\mathcal{D}$	Detection model	D, D'	local/global dataset
$\mathcal{G}$	Local GS set	G	Local GS amount
Γ	Data augmentation radio	N	Time steps for DP-CDM
$\mathcal{R}, \mathcal{R}'$	Total/Partial edge router set	R, R'	Total/Partial edge router amount
T	Communication round	p	L_p norm regularization
$w_r/w_g/W$	DP-CDM weights in edge router / GS / PS	t_g/t_c	Round interval for local update in GS / overall aggregation
ϵ	Privacy budget	δ	Probability of privacy leakage
λ_1, λ_2	Scales for regularization	$\eta_{\mathcal{D}}$	Learning rates for detection model
$\mathcal{L}_{\mathcal{D}}$	Loss of detection model	$\mathcal{L}_w$	Loss of DP-CDM
η_w, η_g, η_o	Learning rates for DP-CDM in local update / local aggregation / overall aggregation	E, E', E''	Local batch epochs for DP-CDM in edge router / local GS / PS
$w_{\mathcal{D}}$	Detection model weights	E^*	Epochs for detection model

A. Dataset Collection

A data collection platform is designed to generate a dataset that simulates STIN traffic distribution. This platform is implemented by OMNET++ [16] and INET [17]. OMNeT++ is a distributed network simulator, whereas INET is its inner model suite for simulating traffic packet configuration and delivery. One scenario is designed to collect benign traffic based on Figure 1 without intruders, whereas extra four attack scenarios are appended in Figure 3 to capture attack traffic.

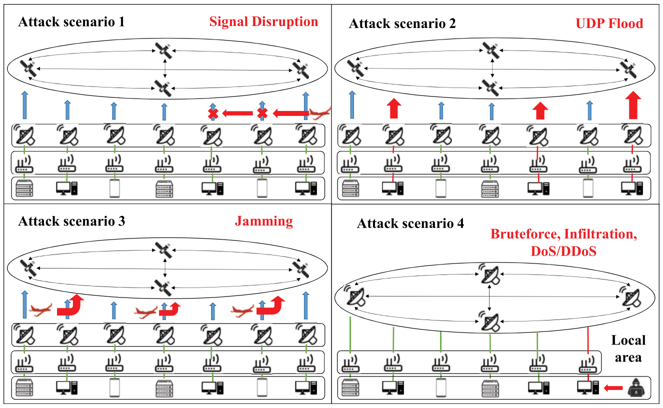


Fig. 3. Four attack scenarios for dataset collection. The first one is the signal disruptions between partial GSs and satellites due to third-party objects; the second one is the UDP flooding launched from terminals to paralyze satellites; the third one is the noise signals with the same radio frequencies sent from third-party objects to directly disturb the satellites; the fourth is the traditional terrestrial attacks based on a local area environment without using satellites.

TABLE III
GENERATED FEATURES IN STI TRAFFIC DATASET

No.	Feature	Category	Description
1	Src_IP	Five-tuple	Source IP address
2	Dest_IP		Destination IP address
3	Src_port		Source port
4	Dest_port		Destination port
5	Protocol		Traffic protocol (UDP, ICMP)
6	Sinr	Intrinsic	Signal-to-interference-plus-noise Radio
7	Duration	Time-based	Transmission duration
8	Inv_mean		Interval time between two packets
9	Inv_min		Minimum time between two packets
10	Inv_max		Maximum time between two packets
11	TTL_min		Minimum time before being discarded
12	TTL_max		Maximum time before being discarded
13	Next_Current_diff		Global time difference between current and next
14	Next_Pre_diff		Global time difference between previous and next
15	SNext_Current_diff		Local time difference between current and next
16	SNext_Pre_diff		Local time difference between previous and next
17	L7_protocol	Content	Application layer protocol
18	DNS_type		DNS query type (A, NS, ...)
19	DNS_TTL_answer		TTL of the first A record (if any)
20	Size	Packet	Packet size or length
21	Throughput		Transmission throughput
22	Flow_bytes_s		Flow bytes per second
23	Flow_packets_s		Flow packets per second
24	DNS_query_id	Random	Transaction ID to a DNS server

The first three scenarios focus on attacks under the satellite domain. Due to a higher level of security in GSs and satellites, most terrestrial intrusions are hard to enter and pollute traffic in the satellite domain [2]. However, signal disruption, jamming, and DoS/DDoS attacks are more changeable and flexible. With similar protocols utilized as STIN benign traffic, these attacks can still affect STIN infrastructures by improving communication losses, interfering with the radio signals, and exhausting target resources. On the other hand, intrusions like Bruteforce and Infiltration are not flexible to intrude on the satellite domain as they depend on the concrete payloads. Their terrestrial versions cannot be applied directly in the satellite domain. Concretely, attack scenario 1 concentrates on the malicious signals that disrupt partial GS-satellite connections. Such attacks are launched by third-party objects, such as drones. As a result, the traffic sent by impacted satellites or GSs possess a huge communication loss rate. Attack scenario 2 generates DDoS attacks from several user terminals to first infect the GSs by sending massive, large and useless packets, then these infected GSs attempt to paralyze satellites based on UDP flooding. Attack scenario 3 contains Jamming attacks that directly interfere with ground-satellite transmission. These attacks are also from third-party objects, escaping the examination from the GSs. The last scenario concentrates on terrestrial intrusions based on a terrestrial environment within a local area. This scenario does not require satellite telecommunication, and is supposed to simulate ground-based attacks including Bruteforce, Infiltration, and DoS/DDoS to intrude the target GSs. Notably, the UDP flooding and terrestrial DoS/DDoS simulated in these scenarios are all based on the bandwidth consumption under the network layer, where the intruders deliberately send massive bogus messages to exhaust the limited bandwidth resources of target infrastructures.

After the scenario establishment, <PcapRecorder> [36] module in OMNeT++ is utilized to record the past traffic in each communication node, whereas <tshark> [37] command is applied to convert <pcap> files into <csv> for training AI-based models. Based on the <PcapRecorder> module, a series of packet features are generated for each captured instance. Due to their characteristics, these features are then classified into six categories, which are five-tuple, intrinsic, content, time-based, packet, and random, respectively. The features and the labels form the new STI dataset, described in Table III.

B. Functional Feature Determination

The functional feature determination process ensures compliance with one constraint in the traffic generation domain: only non-functional features that do not violate the traffic functionality are allowed to be generated [38]. One generated instance $\bar{x}$ by a generative function $f_w(\cdot)$ is described as:

$$\bar{x} = x^{func} \oplus f_w(x^{nonf}) \tag{10}$$

where x^{func} is functional features, x^{nonf} is non-functional features, and $\oplus$ is the concatenation. Applying $f_w(\cdot)$ on x^{func} is not allowed as the traffic functionality may change. Different labels may possess different functional features as different attack types are launched with different means and objectives. These means and objectives make an attack label own its distinct functionality-related features. For example, DoS/DDoS are generated to use multiple causal packets to paralyze targets whereas Infiltration concentrates on achieving concrete objectives with several operations. Consequently, the statistical features like traffic duration and packet sizes have huge values in DoS/DDoS whereas they are generally small in both Infiltration and benign instances. Consequently, these two features are functionality-related for DoS/DDoS but non-functional for Infiltration. Therefore, it is essential to determine the functional and non-functional features of each STI label to achieve functionality-preservation traffic generation. Following the feature categories in Table III, the functional features for different labels are decided in Table IV: five-tuple and intrinsic features reflect the basic traffic packet information, important for all traffic samples; time-based features are highly associated to classify long-time persistent threats like UDP flooding, Jamming, and DoS/DDoS; content-related features

TABLE IV
FUNCTIONAL FEATURE DETERMINATION IN STI DATASET

Labels	Benign	S	U	Jamming	DoS	DDoS	Bruteforce	Infiltration
Five tuple	✓	✓	✓	✓	✓	✓	✓	✓
Intrinsic	✓	✓	✓	✓	✓	✓	✓	✓
Content							✓	✓
Time-based			✓	✓	✓	✓		
Packet		✓	✓	✓	✓	✓	✓	
Random								

S is "Signal Disruption", U is "UDP flood".

are functional for terrestrial Bruteforce and Infiltration as they are related to traffic payload, launched with a fixed objective on subsequent calls; packet-related features are related with the attacks caused by packet loss or massive packet utilization like Signal Disruption and Bruteforce; random feature, the DNS query ID, is non-functional due to its distinctiveness and label independence.

C. Model Training

1) STIN-FL-Based Traffic Generation: Step 3-1 in Figure 2 includes a four-layer FL-based framework to train DP-CDM in a simulated STIN environment. The edge layer includes the edge routers that connect the user terminals. These edge routers act as clients to contribute their calculating power and local recorded traffic. The local layer contains the local GSs for both edge aggregation and higher-level local updates. The satellite layer involves the satellites that work as information transponders. These satellites transmit the model weights between the local GSs and a special GS that is responsible for acting as a PS in the FL-based process. This PS belongs to the global layer for the overall aggregation.

In this step, a new algorithm is designed to train DP-CDM, named STIN-FedAvg in Algorithm 1. Let $\mathcal{G} = \{1, 2, \cdots, G\}$ be the set of local GSs, and each GS, symbolized as g, covers a subset of edge routers $\mathcal{R}_g \in \mathcal{R}$ with $|\mathcal{R}_g| = R_g$. $\mathcal{R}_g$ can be called coverage **g**. Let $\mathcal{G}_r \subseteq \mathcal{G}$ be the local GS subset that edge routers r can communicate with, given $G_r = |\mathcal{G}_r|$. First, the weight w_0 is initialized in each local GS. Second, for each communication round t, R' edge routers download weights w^t from their connected local GSs at line 1 ~ 9. If one edge router r connects with multiple GSs, the downloaded weights are then averaged as:

$$w_r^t = \frac{1}{G_r} \sum_{g \in \mathcal{G}_r} w_g^t \tag{11}$$

Third, the edge routers locally update the weights by their recorded traffic and upload them to local GSs for aggregation at line 10 ~ 12. The details of the local update are illustrated in Algorithm 2, whereas the aggregation in local GSs is:

$$w_g^t = \frac{\eta_g}{R_g} \sum_{r \in \mathcal{R}_g} w_r^t \tag{12}$$

where η_g is the learning rate. Thus, the loss function $f_g(w)$ for coverage **g** is:

$$f_g(w) = \frac{1}{R_g} \sum_{r \in \mathcal{R}_g} F_r(w) \tag{13}$$

where $F_r(w)$ is the loss function of covered edge routers.

Algorithm 1 STIN-FedAvg Training Algorithm

Input: Initialize weight w^0 to each GS g; communication round T; batch epoch $E/E'/E''$; GS local update interval t_g; overall aggregation interval t_c;

1: **for** $t = 1 : T$ **do**
2: Random select R' edge routers $\mathcal{R}'$ from $\mathcal{R}$. $(R' < R)$
3: **for** $g = 1 : G$ **do**
4: **for** $r = 1 : R'$ **do**
5: **if** edge router r is in non-overlapping area **then**
6: $w_r^t = w_g^t$ // Downloading to edge routers
7: **else**
8: $w_r^t = \frac{1}{G_r} \sum_{g \in \mathcal{G}_r} w_g^t$ // Downloading to edge routers
9: **end if**
10: $w_r^t \leftarrow LocalUpdate(E, w_r^t)$ // Algorithm 2 by edge routers
11: $w_g^t = \frac{\eta_g}{R_g} \sum_{r \in \mathcal{R}_g} w_r^t$ // Aggregation to local GSs
12: **end for**
13: **if** $Mod(t, t_g) = 0$ **then**
14: $w_g^t \leftarrow LocalUpdate(E', w_g^t)$ // Algorithm 2 by local GSs
15: **end if**
16: **end for**
17: **if** $Mod(t, t_c) = 0$ **then**
18: $W^t = \frac{\eta_o}{G} \sum_{g \in \mathcal{G}} w_g^t$ // Overall aggregation
19: $W^t \leftarrow LocalUpdate(E'', W^t)$ // Algorithm 2 by PS
20: **for** $g = 1 : G$ **do**
21: $w_g^t = W^t$ // Downloading to local GSs
22: **end for**
23: **end if**
24: $t = t + 1$
25: **end for**

Fourth, for each t_g communication round interval, local GSs also apply their local traffic to train the model at line 13 ~ 16. Finally, for each t_c communication round interval, the weights are uploaded from local GSs to PS through satellites for overall aggregation and highest-level local update with its local traffic at line 17 ~ 19. The overall aggregation in PS is described as:

$$W^t = \frac{1}{G} \sum_{g \in \mathcal{G}} w_g^t \tag{14}$$

while the overall loss function $f(w)$ is described as:

$$f(w) = \frac{1}{G} \sum_{g \in \mathcal{G}} f_g(w) \tag{15}$$

$$= \frac{1}{GR_g} \sum_{g \in \mathcal{G}} \sum_{r \in \mathcal{R}_g} F_r(w) \tag{16}$$

In addition, each local GS downloads the aggregated weight for the next turn at line 20 ~ 22.

After training DP-CDM, PS utilizes this model to synthesize a dataset that contains the global distribution. The details of DP-CDM generation are described in Algorithm 3. Notably, its structure is different from traditional conditional DM. Shown at Figure 4, it includes a functional and non-functional feature division part before the forward process, where this division is dynamic based on the given label. Consequently, for every

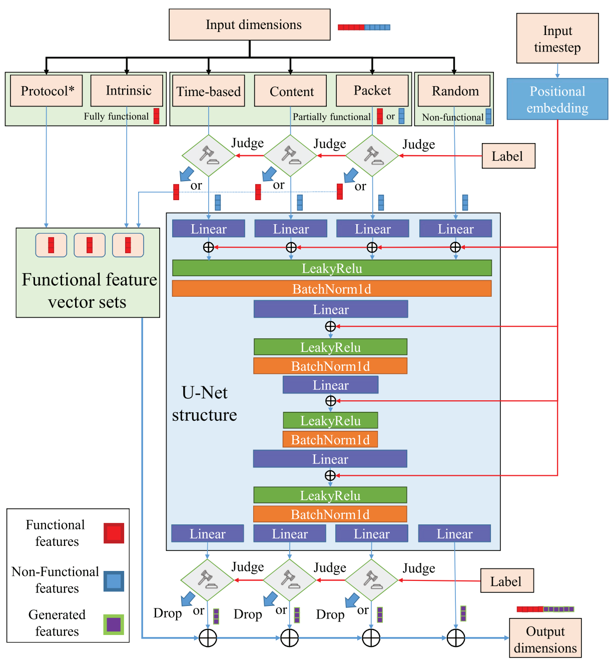


Fig. 4. The structure details of DP-CDM. For each input sample, its features are first divided into functional and non-functional ones. Then, non-functional features are fed into a U-Net-like structure including linear, leaky-relu, batchnorm1d layers together with the concatenation of the timestep positional embeddings for feature generation. Next, the generated features are concatenated with the divided functional features to form a sample. The feature division is dynamic based on the labels throughout the training and generation processes.

input instance, its features are separated into feature categories listed in Table III, which are five-tuple, intrinsic, time-based content, packet and random. Considering that the IP and port features are dropped in the preprocessing step, only the protocol feature in the five-tuple category is retained. Then, these features are judged based on Table IV: the protocol feature and intrinsic category are treated as fully functional, as they are functional for all labels; the time-based, content, and packet categories are treated as partially functional, as the given labels decide whether they are functional or not; the random category is treated as non-functional. Next, its inner neural network is fed by the features judged as non-functional. The network structure is based on U-Net [39], where linear and batchnorm1d layers are applied. To achieve DM-based generation, this network also concatenates the input timestep that is positional embedded after each linear layer. Finally, the output features from this network are connected with the features judged as functional to form a newly generated instance.

This part trains a new DL-based model $\mathcal{D}$ that is supposed to work as NIDS in every edge router, local GS and satellite node. To maximize the detection performance of $\mathcal{D}$, it is trained with both the synthesized dataset and local recorded traffic, where the former represents global STIN traffic distribution while the latter denotes the traffic distribution of the current node. After downloading the synthesized dataset from PS, denoted as D',

[0]Protocol*: IP and port information in the five-tuple feature category are dropped in the preprocessing step. Only protocol feature in five-tuple is used.

Synthetic data-assisted detection model training.

edge router, local GS and satellite nodes train their own $\mathcal{D}$ by both D' and their local data D with the learning rate $\eta_{\mathcal{D}}$. Given $(x, y) \sim D$ and $(x', y') \sim D'$, the objective of $\mathcal{D}$ in each node is minimizing a cross-entropy loss $\mathcal{L}_{\mathcal{D}}$:

$$\mathcal{L}_{\mathcal{D}} = -\frac{1}{M_D}\sum\nolimits_{M_D} y \log \mathcal{D}(x) - \frac{1}{M_{D'}}\sum\nolimits_{M_{D'}} y' \log \mathcal{D}(x') + \Psi \tag{17}$$

where M_D and $M_{D'}$ are the sampled amount from datasets D and D', respectively. We also define $\Gamma = M_{D'}/(M_D + M_{D'})$ to describe the ratio of data augmentation. $\Gamma = 0$ refers that the training data are all from the local data, whereas $\Gamma = 1$ notifies that only D' are applied. Ψ is a L_p−norm regularization for robustness to noisy labels and overfitting alleviation [40]:

$$\Psi = \lambda_1 \left(\sum\nolimits_{M_D} |\mathcal{D}(x)|^p\right)^{1/p} + \lambda_2 \left(\sum\nolimits_{M_{D'}} |\mathcal{D}(x')|^p\right)^{1/p} \tag{18}$$

where λ_1 and λ_2 are two predefined hyperparameters.

D. Online Detection

The trained $\mathcal{D}$ is then deployed in each edge router, GS and satellite to work as a part of NIDS for online detection in STIN. Deployment on these nodes can reflect the whole flow circumstance of STIN as these nodes are indeed traffic pivots in STIN. For each node, when traffic packets come in, they are first checked by a firewall filter, which can directly block obvious attacks with a set of defined rules. Then, the passed packets are detected by a NIDS module consisting of a flow-based feature collector, a preprocessing part, and $\mathcal{D}$. If this module found an abnormal packet, it would raise an alert and launch follow-up analysis and mitigation modules based on the classified attack types. Otherwise, the packet is dropped.

Additionally, the satellite nodes may be restricted with computing resources, but they just participate in training and prediction processes of $\mathcal{D}$ instead of DP-CDM, which is less time-consuming. Moreover, the training process of $\mathcal{D}$ is implemented only once per t_c communication rounds. In most cases, these traffic pivots only execute intrusion detection tasks. Therefore, the resource consumption on these satellite nodes is indeed acceptable.

V. Analysis of the Proposed Framework

The proposed STINIDF is analyzed in security and time complexity theoretically in this section.

A. Security Analysis

This part analyzes the potential threats that affect the STIN-based training process in STINIDF and its privacy issues, including inference attacks, injection attacks, and data privacy.

1) Inference Attack on Edge Router: Considering that GSs and satellites are configured with higher level safety protection, the intruders may apply an inference attack on any edge routers for deducing the original traffic data based on the weight and gradient information of DP-CDM. To against it, DP is utilized for local updating where Gaussian noise $\mathcal{N}(0, \sigma^2 C^2 \mathrm{I})$ in line 10 of Algorithm 2 is added to each gradient. Meanwhile, DP-CDM is only fed with non-functional traffic features, not highly associated with the functionality of raw data. The functional features do not participate in the federated training. Even if the intruders successfully deduce the obtained weight and gradient, the original traffic instances can not be completely reconstructed due to the lack of their functional elements. Thus, STINIDF is secure when facing inference attacks.

2) Injection Attack for Model Corruption: The intruders may infect and control several edge routers to launch injection attacks by uploading malicious weight information. This may lower the performance of the global aggregated DP-CDM and make it unreliable. STINIDF alleviates this risk mainly based on multi-level aggregation/local updating, where the local GSs instead of PS aggregate the uploaded weights in advance. Then these GSs utilize their local traffic to further update the aggregated weights before uploading to PS. Moreover, the recorded traffic in local GSs is generally more massive and informative, lessening the training participation degree of the edge routers. Conclusively, the model corruption risk is significantly reduced because of the multi-level factors in STINIDF.

3) Data Privacy for Local Raw Traffic: In the federated process of step 3-1, each edge router or local GS applies its own recorded raw traffic to train individual DP-CDM, which is then uploaded for weight aggregation. In the detection model training process of step 3-2, only the dataset generated by PS is downloaded by each node for training $\mathcal{D}$. Overall, the raw traffic recorded by each node does not exchange during STINIDF training process. Hence, its privacy can be guaranteed.

B. Time Complexity Analysis

The computation and communication costs of STINIDF are given for each edge router, satellite, local GS and PS.

1) Computation Cost: For edge routers, they include two parts: 1) locally updating DP-CDM in every communication round, the cost is $\mathcal{O}(E(B|w_r|+l_{dp}))$, where clipping the sample gradients and adding noise into gradient in Algorithm 2 require $\mathcal{O}(l_{dp})$ operations; 2) training $\mathcal{D}$ once per t_c communication rounds, the cost is $\mathcal{O}(E^*((M_{D'} + M_D)|w_{\mathcal{D}}| + l_{re}))$, where l_{re} denotes consumption of Ψ. Therefore, the total computation consumption for one edge router is $\mathcal{O}(T(E(B|w_r| + l_{dp}) + E^*((M_{D'} + M_D)|w_{\mathcal{D}}| + l_{re})))$. For each satellite node, it only trains $\mathcal{D}$ after downloading the synthetic dataset per t_c communication rounds. Its cost is $\mathcal{O}(T(E^*((M_{D'}+M_D)|w_{\mathcal{D}}|+l_{re})))$. For local GSs, they include three parts: 1) locally updating DP-CDM per t_g communication rounds, the cost is $\mathcal{O}(E'(B|w_g| + l_{dp}))$; 2) aggregating the uploaded weights from edge routers per t_g communication rounds, the cost is $\mathcal{O}(R'|w_r|)$; 3) training $\mathcal{D}$ once per t_c communication rounds. Thus, the total computation consumption for single local GS is $\mathcal{O}(T(E'(B|w_g|+l_{dp})+R'|w_r|+E^*((M_{D'}+M_D)|w_{\mathcal{D}}|+l_{re})))$. For PS, it includes three parts: 1) local updating DP-CDM per t_c communication rounds, the cost is $\mathcal{O}(E''(B|W| + l_{dp}))$; 2) aggregating the uploaded weights from local GSs per t_c communication rounds, the cost is $\mathcal{O}(G|w_g|)$; 3) generating the synthetic dataset by aggregated DP-CDM, the cost is $\mathcal{O}(N|D'||W|)$. Hence, the total computation consumption for PS is $\mathcal{O}(T(E''(B|W| + l_{dp}) + G|w_g| + N|D'||W|))$.

2) Communication Cost: For edge routers, 1) R' of them upload updated DP-CDM in every communication round, the cost is $\mathcal{O}(|w_r|)$; 2) they download the synthetic dataset after global aggregation, the cost is $\mathcal{O}(|D'|)$. Therefore, the communication cost for one edge router is $\mathcal{O}(T(|w_r| + |D'|))$. For each satellite, it only downloads the synthetic dataset, the cost is $\mathcal{O}(T|D'|)$. For local GSs, 1) they receive uploaded DP-CDM weights from edge routers per t_g communication rounds, the cost is $\mathcal{O}(R'|w_r|)$; 2) they upload updated DP-CDM weights to PS, the cost is $\mathcal{O}(|w_g|)$; 3) they download the synthetic dataset from PS, the cost is $\mathcal{O}(|D'|)$; 4) they forward the synthetic dataset to each edge router, the cost is $\mathcal{O}(R'|D'|)$. Thus, the communication cost for local GS is $\mathcal{O}(T(R'|w_r| + |w_g| + (R' + 1)|D'|))$. For PS, 1) it receives DP-CDM weight from local GSs, the cost is $\mathcal{O}(G|w_g|)$; 2) it sends the synthetic dataset to each local GS, the cost is $\mathcal{O}(G|D'|)$. Hence, its total communication cost is $\mathcal{O}(T(G|w_g| + |W| + G|D'|))$.

Meanwhile, the complexity and scale of $\mathcal{D}$ are generally much less than DP-CDM, leading that the computational term $E^*((M_{D'} + M_D)|w_{\mathcal{D}}| + l_{re})$ in edge router and local GS nodes can be omitted. Also, the size of DP-CDM in different nodes is the same (i.e., $|w_r| = |w_g| = |W|$). Consequently, all costs are simplified and concluded in Table V. The costs of FedAvg [9] and CIDIoT [33] are recomputed for the STIN scenario and listed as references. Conclusively, the costs of STINIDF mainly depend on the size of DP-CDM (i.e., $|W|$) and the amount of the synthetic dataset (i.e., $|D'|$). As for $|W|$, its size is based on the inner U-Net. Compared with the federated DM work for image generation [41] and federated U-Net works for medical segmentation [42], [43], the scalability of this U-Net is indeed acceptable in communication contexts as it generates traffic instances mainly by linear and batchnorm1d layers, whose parameters are much fewer than convolutional and batchnorm2d layers in image-related usage.

It is worth noting that STINIDF is indeed designed for high intrusion detection performance in STIN instead of pursuing low time consumption. Based on Table V, it has paid extra resources on achieving multi-level aggregation/local-updating for smoothing the federated process and DP operations for weight privacy protection compared with other FLs. Nevertheless, the extra resource consumption is acceptable. Because the local GSs, commonly configured with sufficient computing resources, play the main role of multi-level aggregation/local-updating, whereas DP operations are applied only on the local data, whose amount is limited in single node like edge routers.

VI. Dataset Collection and Experiment Design

This section describes the collected STI dataset, followed by the experiment setups for the proposed STINIDF.

A. STI Dataset Formulation

OMNeT++5.6.2 and INET 4.2.5 are utilized to achieve STI dataset formulation. Based on the different attack scenarios in Figure 3, seven attack labels under terrestrial and satellite domains are collected. OMNeT++ has spent 3430 seconds to

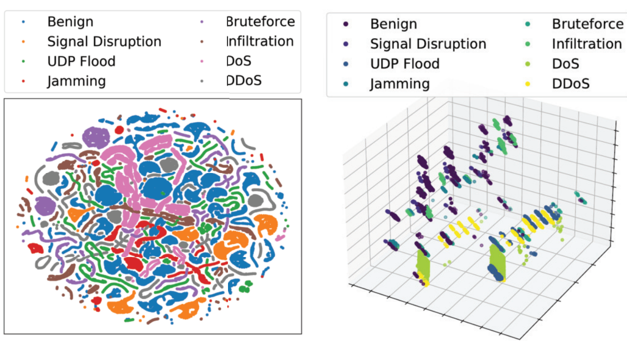
TABLE V
COMPUTATION AND COMMUNICATION COSTS OF RELATED METHODS

Method	Multi	DP	Node	Computation cost	Communication cost
FedAvg [9]			Client	$\mathcal{O}(TEB\lvert W\rvert)$	$\mathcal{O}(T\lvert W\rvert)$
			PS	$\mathcal{O}(TR'\lvert W\rvert)$	$\mathcal{O}(TR'\lvert W\rvert)$
CIDIoT [33]		✓	Edge node	$\mathcal{O}(T\tau^2 + TE(B\lvert W\rvert + l_{dp}))$	$\mathcal{O}(2T\lvert W\rvert + Tl_{ss})$
			Cloud	$\mathcal{O}(T(2EB\lvert W\rvert + \tau n))$	$\mathcal{O}(2T\lvert W\rvert + TR'l_{ss})$
STINIDF	✓	✓	Edge router	$\approx \mathcal{O}(T(E(B\lvert W\rvert + l_{dp})))$	$\mathcal{O}(T(\lvert W\rvert + \lvert D'\rvert))$
			Satellite	$\mathcal{O}(T(E^*((M_{D'} + M_D)\lvert w_{\mathcal{D}}\rvert + l_{re})))$	$\mathcal{O}(T\lvert D'\rvert)$
			Local GS	$\approx \mathcal{O}(T(E'(B\lvert W\rvert + l_{dp}) + R'\lvert W\rvert))$	$\mathcal{O}(T(R'+1)(\lvert W\rvert + \lvert D'\rvert))$
			PS	$\mathcal{O}(T(E''(B\lvert W\rvert + l_{dp}) + (N\lvert D'\rvert + G)\lvert W\rvert))$	$\mathcal{O}(T((G+1)\lvert W\rvert + G\lvert D'\rvert))$

"Multi" refers multi-level aggregation/local-updating usage, "DP" refers DP usage,
τ, n obey (τ, n) threshold secret-sharing method [44], whereas $\mathcal{O}(l_{ss})$ refers the secret sharing consumption in CIDIoT.

TABLE VI
LABEL SIMULATION TIME AND AMOUNT IN STI TRAFFIC DATASET

Domain	Label	Simulation time	Amount
-	Benign	$0 - 1000$s	678525
Satellite	Signal Disruption	$1000 - 1230$s	154263
	UDP flood	$1230 - 1670$s	245582
	Jamming	$1670 - 1910$s	143191
Terrestrial	Bruteforce	$1910 - 2190$s	195344
	Infiltration	$2190 - 2460$s	182572
	DoS	$2460 - 2810$s	248256
	DDoS	$2810 - 3230$s	274589



(a) 2-dimensional T-SNE (b) 3-dimensional PCA

Fig. 5. Low-dimensional visualization for partial STI dataset. They show that the label distributions in STI present clear differences in statistical level.

simulate benign and malicious traffic. The information about this simulation process are recorded in Table VI.

Then this dataset is pre-processed by feature selection and normalization. The IP and port information features are treated as redundant features and dropped. For features that vary sharply, a min-max normalization is applied to scale them within a range $[0, 1]$. Then, the normalized STI dataset with 20 features are obtained, public at https://github.com/hjp007/STI.

Moreover, this pre-processed dataset is visualized by 2-dimensional T-SNE [45] in Figure 5(a) and 3-dimensional PCA [46] in Figure 5(b) with partial samples, illustrating that each STI malicious traffic label possesses a difference to some extent in statistical distribution compared with other labels.

B. Experiment Setups

The experiment setups include environment settings, a default node scenario configuration, hyper-parameter details, evaluation metrics, and baseline settings.

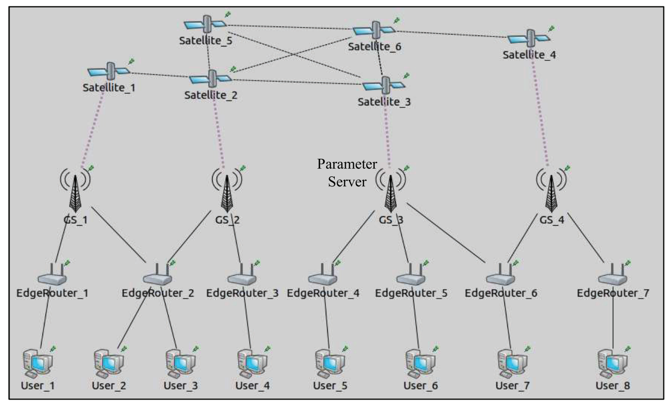


Fig. 6. Node scenario setting for simulating FL-based training in a STIN environment. GS_3 is the PS that is responsible for overall aggregation for DP-CDM and synthesizing global dataset.

1) Experimental Environment: A computer with Intel Core i9-10700K CPU, 32 GB DDR3, and GeForce RTX 3060Ti GPU is utilized. Jupyter Notebook 7.1.3 platform is applied to train and test the related DL-based models. These models are programmed by PyTorch 1.13.1 with Python 3.10.8.

2) Default Node Scenario Configuration: To train DP-CDM in step 3-1 in Figure 2, a default node scenario is pre-defined to simulate the STIN environment, shown in Figure 6. It includes 6 satellite nodes, 4 GS nodes, 7 edge routers, and 8 user terminals. GS_3 is set as PS, and other GS nodes are local GSs. Thus, G is 3 and R is 7. Considering the overlapping downloading factor, one edge router may connect multiple GSs (i.e., EdgeRouter_2 and EdgeRouter_6). One edge router may also connect multiple users (i.e., EdgeRouter_2). Meanwhile, the data distribution in each node that participates in Algorithm 1 (i.e., all edge routers and GSs) is set into two conditions, which are IID and non-IID scenarios. The IID scenario requires that the malicious traffic in the satellite domain is averagely distributed in GS nodes, while the terrestrial attack samples are equally scattered in edge routers and GSs. The non-IID scenario refers to the fact that the distributions in both domains are skewed and imbalanced. In both scenarios, the satellite domain traffic never exists on edge routers.

3) Hyper-Parameter Details: The STI dataset is divided into a 70% training set and a 30% testing set. The training set is utilized to train DP-CDM where the communication round T is set as 30 and the batch epochs $E/E'/E''$ are 5, 5, 5. Moreover, the batch size B, the partial edge router amount

R', the local update interval t_g, and the overall aggregation interval t_c are set as $32, 3, 3, 5$, respectively. The learning rates η_w, η_g, η_o are all 0.001. In the structure of DP-CDM, the time step N is 50 and β_n changes from 0.001 to 0.02 in these N steps. To evaluate DP effectiveness in DP-CDM, subsection VII-E studies the performance impact under the different privacy budgets ϵ, where ϵ is set as $0.1, 1, 10, \infty$, respectively and the risk probability δ is 0.001 ($\epsilon = \infty$ refers no DP usage).

For each edge router, local GS, and satellite, its deployed detection model $\mathcal{D}$ is trained by the downloaded synthetic dataset from PS and local training dataset for every overall aggregation round interval t_c with 20 epochs (i.e., $E^* = 20$), then evaluated by the local testing dataset. Learning rate $\eta_{\mathcal{D}}$ is 0.001. As for regularization terms, we chose L_2-norm where $p = 2$, the hyperparameters λ_1, λ_2 are both set as 0.1. The data augmentation ratio Γ is set as 0.75 for the IID scenario and 0.25 for the non-IID scenario, as observed in subsection VII-C. Besides, the structure of $\mathcal{D}$ uses a linear deep neural network, including four fully-connected layers, three batchnorm1d layers, and one softmax output layer to achieve intrusion multi-classification.

4) Evaluation Metrics: Based on true positive (TP), true negative (TN), false positive (FP), and false negative (FN), the following evaluation metrics are selected to measure the detection performance of trained $\mathcal{D}$ and other methods:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FN} + \text{FP}} \tag{19}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{20}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{21}$$

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{22}$$

To evaluate the generated instances, FID [47] is adopted, where a smaller value implies better instance quality. FID utilization requires a pre-trained calculating workflow from our previous work [38] to adapt the domain of traffic generation.

5) Baseline Methods: Several state-of-the-art studies based on FL are implemented for performance comparison. Some of them are not designed for the intrusion detection domain but are retrained over the STI dataset in our experiments.

- FedAvg [9]: the basic FL that averages the model weights from the local clients to achieve global model training.
- FedProx [48]: a FL based on FedAvg that adds a proximal term to mitigate the statistical data heterogeneity.
- SDA-FL [12]: a FL with confident pseudo labeling to solve the non-IID challenge by sharing the synthetic data from GAN. DP is also utilized for privacy protection.
- FedDPGAN [21]: a FL that privately generates diverse instances by GAN in local clients for diagnosing COVID-19. DP is also utilized for privacy protection.
- CIDIoT [33]: a FL with a loss sensitive GAN and a robust aggregation process for IoT-environment-adapted intrusion detection. DP is also utilized for privacy protection.
- STIN-NIDS [2]: a distributed FL-adapted NIDS to properly allocate resources in NIDS domain to analyze and block malicious traffic with complexity optimization.

Meanwhile, random forest (RF) [49] and multi-layer perceptron (MLP) [50] are also adopted to play the role of $\mathcal{D}$, trained with local data in each edge router, local GS, and satellite.

VII. Experiment Results and Discussions

The evaluation results for STINIDF are categorized into seven key aspects. First, a direct comparison with SOTA methods is provided. Second, the concrete classification performance of STINIDF on each label is examined. Third, results under varying Γ are presented to identify the optimal data augmentation degree. Fourth, the evaluation for different node types is conducted. Fifth, the impact of varying privacy budget settings is examined to determine the optimal trade-off for DP usage. The sixth and seventh aspects encompass a scalability study and an ablation study. Each aspect is accompanied by a corresponding discussion. Finally, the limitations of STINIDF are critically addressed.

A. Comparison With SOTA

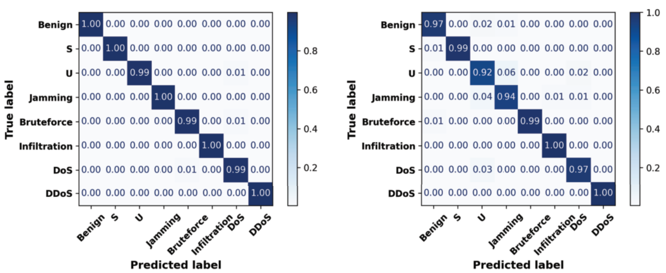
The proposed STINIDF is evaluated against the aforementioned baseline methods on detection performance in both IID scenario and non-IID scenario. The experiment is conducted five times for every method to get the average accuracy, precision, recall and F1 score, along with their standard deviations. The results, summarized in Table VII, clearly demonstrate that STINIDF gets the best performance among all methods in both scenarios across these four metrics. It achieves performance improvements compared with the second-best method, which are 0.2% accuracy, 0.13% precision, 0.38% recall, 0.24% F1 score in the IID scenario, and 2.41% accuracy, 3.14% precision, 1.65% recall, and 2.7% F1 in the non-IID scenario.

Discussion: Non-FL-based methods, such as RF and MLP, exhibit poor performance in both scenarios. Without sharing the model weights that contain global traffic information, these models are trained solely with limited local data, resulting in incomplete convergence and instability. Consequently, their performance shows significantly higher deviations (e.g., MLP in the non-IID scenario even gets 24.51% deviations in precision). For FL-based methods that are not designed for solving the non-IID challenge, FedAvg and STIN-NIDS, they reveal more evident performance degradation from the IID scenario to the non-IID scenario than others. For instance, FedAvg and STIN-NIDS get 7.22%, 10.85% lower accuracy whereas 3.04% in STINIDF. This illustrates that the strategies like data augmentation or adding the proximal term do improve the detection performance in the non-IID scenario.

Meanwhile, STINIDF outperforms FedProx, SDA-FL, FedDPGAN and CIDIoT with 96.63% accuracy, 96.71% precision, 96.54% recall and 96.66% F1 score in the non-IID scenario. It shows that the data augmentation strategy utilized in STINIDF has balanced the distribution of the local traffic in each deployed node, whose $\mathcal{D}$ can then effectively detect its passed intrusions. In conclusion, STINIDF is able to model the traffic representations inherent in STI dataset under unbalanced data scenario and outperforms other SOTA methods.

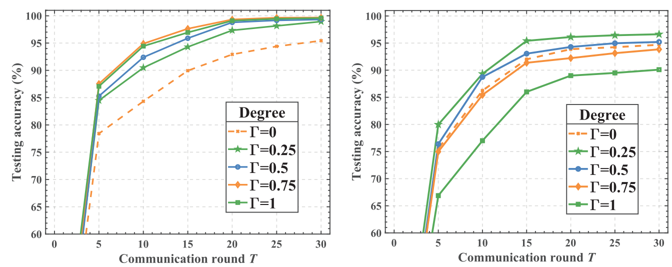
TABLE VII
OVERALL TESTING ACCURACY, PRECISION, RECALL AND F1 SCORE WITH COMPARED STATE-OF-THE-ART METHODS (%)

Scenarios	IID scenario				Non-IID scenario			
Algorithm	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
FedAvg [9]	94.45 ± 0.30	94.32 ± 0.29	94.51 ± 0.30	94.44 ± 0.29	87.23 ± 2.22	86.90 ± 1.59	86.65 ± 3.77	87.17 ± 1.91
FedProx [48]	95.54 ± 0.82	94.97 ± 1.02	95.8 ± 0.46	95.54 ± 0.49	92.37 ± 1.15	92.41 ± 0.85	92.33 ± 1.2	92.4 ± 0.79
SDA-FL [12]	99.18 ± 0.57	99.16 ± 0.61	99.19 ± 0.48	99.18 ± 0.3	94.22 ± 2.32	93.57 ± 0.63	94.89 ± 1.11	93.96 ± 1.42
FedDPGAN [21]	96.54 ± 1.15	95.23 ± 1.04	98.14 ± 0.89	96.68 ± 1.01	91.42 ± 3.33	92.57 ± 3.14	88.87 ± 1.37	90.44 ± 2.86
CIDIoT [33]	99.47 ± 0.43	99.51 ± 0.39	99.33 ± 0.38	99.44 ± 0.28	93.1 ± 2.14	91.56 ± 3.15	94.89 ± 0.97	92.83 ± 1.58
STIN-NIDS [2]	98.14 ± 0.25	98.21 ± 0.1	98.01 ± 0.49	98.14 ± 0.17	87.29 ± 3.24	87.91 ± 4.44	86.37 ± 3.6	86.81 ± 3.05
RF	91.89 ± 0.84	91.60 ± 0.32	92.04 ± 0.24	91.84 ± 0.15	79.43 ± 6.25	81.24 ± 3.88	76.1 ± 4.32	77.66 ± 4.14
MLP	89.32 ± 1.53	90.18 ± 0.45	88.22 ± 2.33	88.97 ± 1.15	68.49 ± 12.68	56.54 ± 24.51	69.36 ± 8.39	66.58 ± 10.08
STINIDF	$\mathbf{99.67 \pm 0.25}$	$\mathbf{99.64 \pm 0.32}$	$\mathbf{99.71 \pm 0.29}$	$\mathbf{99.68 \pm 0.14}$	$\mathbf{96.63 \pm 1.15}$	$\mathbf{96.71 \pm 1.23}$	$\mathbf{96.54 \pm 0.84}$	$\mathbf{96.66 \pm 0.97}$



(a) IID scenario (b) Non-IID scenario

Fig. 7. Confusion matrices of STINIDF on STI dataset in both IID and non-IID scenarios. S is "Signal Disruption", U is "UDP flood".



(a) IID scenario (b) Non-IID scenario

Fig. 8. Overall testing accuracy for STINIDF under different data augmentation ratio Γ. Γ = 0 refers no global synthetic dataset (i.e., no data augmentation usage) while Γ = 1 refers training without local dataset.

B. Classification Performance for Intrusion Detection

To further assess the classification performance of the proposed method, confusion matrices detailing the results on each label are displayed in Figure 7, with evaluations conducted under both IID and non-IID scenarios.

Discussion: In the IID scenario, STINIDF achieves over 99% accuracy on all labels. In the non-IID scenario, STINIDF exhibits slight performance degradation on Benign, UDP Flood, Jamming, and DoS traffic. The accuracy is reduced by approximately 3%, 7%, 6%, 2%, respectively. A plausible explanation for this lies in the shared functional feature components, time-based and packet-related features, among UDP Flood, Jamming, and DoS based on Table IV. These similarities result in closely resembling characteristics to each other. When the training process becomes sophisticated like the non-IID scenario, the classification task on them becomes tough in the first place, causing a declined detection result. Besides, the performance on benign instances is also declined as their distribution is more diverse in feature space, shown in Figure 5. This diversity causes some benign instances to share characteristics with certain attack instances, further complicating classification.

C. Data Augmentation Degree

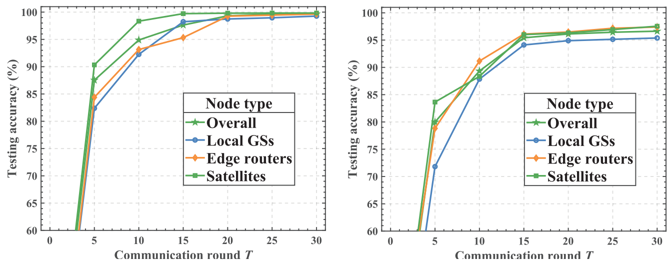
For synthetic data-assisted detection model training, it is essential to verify the effectiveness of the data augmentation and figure out an appropriate degree. For STINIDF, this factor is highly correlated to the data augmentation ratio Γ. To find the best Γ, it is set into $0, 0.25, 0.5, 0.75, 1$ for both IID and non-IID scenarios. $\Gamma = 0$ refers no data augmentation usage whereas $\Gamma = 1$ implies no local data usage. Figure 8 shows the overall testing accuracy during each communication round T under these Γ. As a consequence, STINIDF gets the best performance when $T = 30$ in different Γ settings. Concretely, it achieves 95.43%, 98.94%, 99.32%, 99.67%, 99.58% overall testing accuracy in the IID scenario and 94.69%, 96.63%, 95.21%, 93.85%, 90.1% in the non-IID scenario.

Discussion: In Figure 8, STINIDF possesses the best performance 99.67% when $\Gamma = 0.75$ in the IID scenario and 96.63% when $\Gamma = 0.25$ in the non-IID scenario.

In the IID scenario, $\Gamma = 0$ achieves 95.43% accuracy, significantly lower than other Γ conditions. One reason is the lack of training instances. $\Gamma = 0$ can only utilize local training data, whereas other Γ conditions can download the generated instances that contain the traffic information from other nodes. Meanwhile, $\Gamma = 0.75$ performs slightly better than $\Gamma = 1$. It shows that the local data is still required as it represents the original data distribution.

In the non-IID scenario, $\Gamma = 1$ gets 90.1% accuracy, even worse than $\Gamma = 0$. One reason is that only the local training set, instead of the global synthetic dataset, shares the same data distribution as the local testing set in this scenario. Thus, local training data is more important than global data for $\mathcal{D}$. On the other hand, $\Gamma = 0.25, 0.5$ still works better than $\Gamma = 0$, illustrating that data augmentation with an appropriate degree can obtain a higher accuracy than no data augmentation.

Conclusively, STINIDF utilizes a data augmentation strategy by applying the synthetic dataset symbolizing global traffic distribution without dropping local data in each deployed node. The synthetic dataset contributes to $\mathcal{D}$ with stronger



(a) IID scenario (b) Non-IID scenario

Fig. 9. Average testing accuracy on each type of nodes and overall condition for STINIDF. The performance is different because different types of nodes possess distinctive local datasets in different domains.

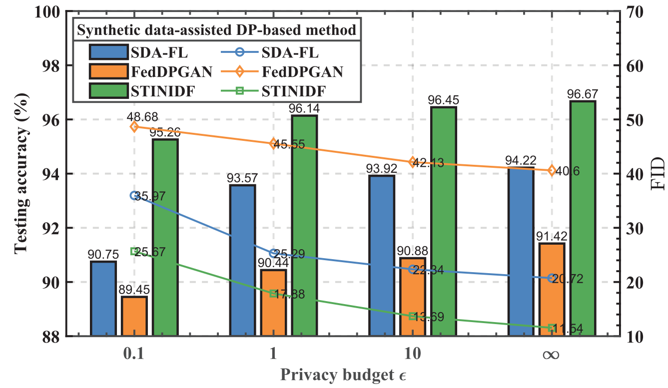


Fig. 10. Overall testing accuracy and FID under different privacy budgets ϵ by STINIDF and other methods in non-IID scenario. $\epsilon = \infty$ refers to no usage of differential privacy, while smaller ϵ symbolizes stronger privacy protection.

robustness and generalization, whereas local data guarantees $\mathcal{D}$ to maintain prediction capability for its current node traffic distribution. The results have confirmed that this strategy is effective for $\mathcal{D}$ to obtain higher performance in both scenarios.

D. Detection Performance in Each Node Types

Three types of nodes, edge routers, local GSs and satellites, host $\mathcal{D}$ for final online detection, as illustrated in Figure 2. Given that $\mathcal{D}$ in different node types is trained with distinct local data distributions, an experiment is implemented to inspect the performance differences in these node types. The results, summarized in Figure 9, show that $\mathcal{D}$ achieves 99.27%, 99.64%, 99.83%, 99.67% on local GSs, edge routers, satellites and overall in the IID scenario, while 95.38%, 97.39%, 97.54%, 96.63% in the non-IID scenario.

Discussion: In both IID and non-IID scenarios, the performance of $\mathcal{D}$ on local GS is slightly lower than that on the other two node types. This can be attributed to the complexity of the traffic composition. The local malicious data in the satellites and edge routers is from a single domain, i.e., the satellite or the terrestrial domain in Figure VI. Besides, the local data in the local GS is from both domains, as the local GS plays the role of the convergence center, handling all kinds of traffic in the STIN environment. $\mathcal{D}$ in the local GS is required to classify all the labels in both domains, which increases complexity and slightly reduces performance.

Meanwhile, $\mathcal{D}$ in the satellite node slightly outperforms $\mathcal{D}$ in the edge router node, indicating that the traffic complexity in the terrestrial domain is marginally higher than the satellite domain. One possible interpretation, based on Table IV, is that the functionality of the malicious traffic in the satellite domain is correlated with the packet or time features, whereas the terrestrial intrusions also have a relationship to the content-related features, leading to a harder classification task for $\mathcal{D}$ to detect the terrestrial intrusions.

E. Trade-off Between the Privacy Budgets and Performance

To prevent privacy leakage during weight exchange in Algorithm 1 for DP-CDM, (ϵ, δ)-DP is utilized where privacy budget ϵ is set into several values, including ∞, to determine an optimal balance for final performance. $\epsilon = \infty$ refers no DP usage. The smaller ϵ indicates stronger privacy protection but huger fuzziness in the exchanged weight. The final performance is assessed based on the quality of the instances generated by DP-CDM and the detection capability of the augmented $\mathcal{D}$. Besides, SDA-FL and FedDPGAN are chosen as references as they also apply data augmentation and DP. Consequently, Figure 10 presents corresponding FIDs and overall testing accuracy under these ϵ in the non-IID scenario. Basically, for different ϵ settings, STINIDF achieves the highest testing accuracy set (95.26% ~ 96.67%) whereas it generates the instances with the lowest FID set (25.67 ~ 11.54) compared to the other methods.

Discussion: Compared to the absence of DP, a smaller ϵ increases the noise scale σ, thereby expanding the variance of the added noise $\mathcal{N}(0, \sigma^2 C^2 \mathrm{I})$ to DP-CDM in line 10 of Algorithm 2. While this enhances privacy protection during weight exchange in Algorithm 1, it makes trained DP-CDM with slight deviation, causing the generated instances with lower quality by 14.13, 6.34, 2.15 higher FIDs. Naturally, training $\mathcal{D}$ with these instances decreases its performance by 1.41%, 0.53%, 0.22% lower accuracy on original traffic data.

Notably, this performance degradation compared with no DP usage is not significant when $\epsilon = 1, 10$ (i.e., 0.53%, 0.22%), whereas it is more evident when $\epsilon = 0.1$ (i.e., 1.41%). Therefore, $\epsilon = 1$ is a suitable trade-off between privacy protection and model performance.

F. Scalability Study

To investigate the impact of varying number of local GSs and edge routers on STINIDF performance, additional G and R values are examined beyond the default node configuration (i.e., $G = 3$, $R = 7$). By the experiment, the final testing accuracy with G of $0 \sim 4$ and R of $1 \sim 9$ in the non-IID scenario is examined and recorded in Figure 11. $G = 0$ symbolizes single PS usage with no local GS setting.

Discussion: First, the model possesses significantly higher accuracy (i.e., 98.1 ~ 99.71%) when $G = 0$ compared with $G = 1 \sim 4$. This is because no local GS exists, which is a special condition. It causes the model training process in Algorithm 1 to degrade into a single-level many-to-one aggregation (i.e., the edge routers are directly connected to PS), lowering the training difficulty. Second, the accuracy drops with larger

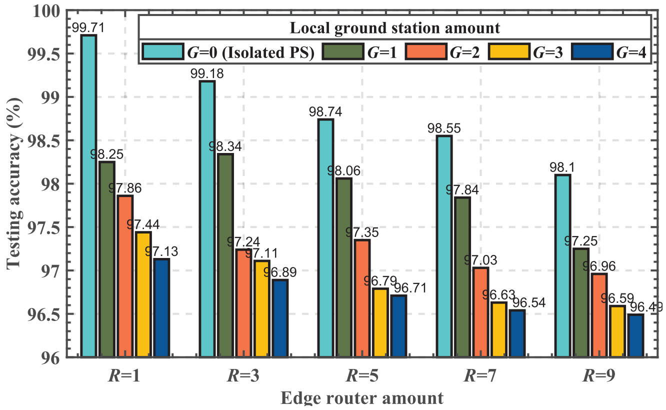


Fig. 11. Overall testing accuracy under different numbers of local GSs (G) and edge routers R in the non-IID scenario. Generally, larger G and R cause a lower accuracy, while this trend becomes slower when $G \geq 3$ and $R \geq 7$.

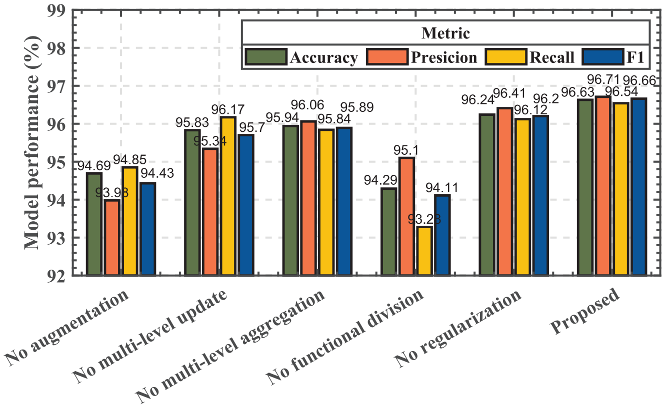


Fig. 12. Overall accuracy, precision, recall and F1 score under different ablation conditions in the non-IID scenario, showing that the data augmentation, the multi-level update/aggregation, the functional feature division, and the regularization do improve the performance.

G and R, indicating that the node scenario with a larger scale lowers the aggregation and final performance. The reason is that more G and R make the local traffic instances stored in each STIN node become fewer under the fixed amount of STI dataset. Consequently, the larger scale of STIN nodes without appending total training instances inevitably impedes the performance of FL-based aggregation. Moreover, fewer local instances in each STIN node make $\mathcal{D}$ hard to learn the traffic distribution in its own node. However, in a real-world scenario, each STIN node is supposed to collect its passed traffic, which is then stored as its local data. In this case, a larger scale of STIN nodes, together with informative local instances may not lower the aggregation and final performance. Third, the performance decreasing degree between $R = 7$ and $R = 9$ (i.e., 0.45% ↓, 0.59% ↓, 0.07% ↓, 0.08% ↓, 0.05% ↓) is roughly smaller than the one between $R = 3$ and $R = 5$ (i.e., 0.44% ↓, 0.28% ↓, 0.11% ↑, 0.32% ↓, 0.18% ↓). This is the same for G variation, where the decreasing degree between $G = 3$ and $G = 4$ (i.e., 0.31% ↓, 0.22% ↓, 0.08% ↓, 0.13% ↓, 0.1% ↓) is generally smaller than the one between $G = 1$ and $G = 2$ (i.e., 0.39% ↓, 1.1% ↓, 0.71% ↓, 0.71% ↓, 0.29% ↓). It illustrates that the impact of the node amount setting becomes weaker with the scale of the STIN going up. In reality, the deployed STIN systems are generally with thousands of edge routers and hundreds of GSs and satellites. Thus, the impact of small node variation numbers on these systems can be ignored.

G. Ablation Study

To access the effectiveness of the independent modules in STINIDF, an ablation experiment is implemented in the non-IID scenario, examining five conditions: no data augmentation usage (i.e., $\Gamma = 0$), no multi-level update/aggregation in Algorithm 1, no functional feature division in Table IV, and no regularization (i.e., no Ψ in Eq (17)). The accuracy, precision, recall and F1 score on these conditions compared to the original method are collected in Figure 12. Consequently, the proposed STINIDF outperforms all conditions and gets 1.94%, 0.8%, 0.69%, 2.34%, 0.39% higher accuracy, respectively.

Discussion: First, compared with no data augmentation usage (i.e., no DP-CDM usage), the absence of multi-level updates in DP-CDM training improves 1.14% accuracy, whereas the absence of multi-level aggregation improves 1.25% accuracy. This indicates that both multi-level characteristics enhance performance, while the multi-level update possesses a slightly larger degree. One reason is that the multi-level update does utilize more local data (i.e., the data in local GSs), which is more important than the optimization of the aggregation process. Second, omitting functional feature division results in the worst performance across four metrics, even worse than no data augmentation. On the one hand, no division step before traffic generation is illegal, as the generated traffic may violate its functionalities due to the functional feature modification. On the other hand, no division may let DP-CDM generate instances with unseen functional features. Compared with the ones with unmodified functional features, these instances are more diverse and may be misleading for training $\mathcal{D}$, as the performance of $\mathcal{D}$ is evaluated by the original test data with unmodified functional features. Meanwhile, the amount of features that require generation also improves, causing a harder DP-CDM training process. Third, no regularization gets 0.39% lower accuracy, indicating that regularization does assign $\mathcal{D}$ with a certain degree of generalization capability improvement.

H. Limitations

Three limitations exist in STINIDF, which are the potential leakage of the synthetic data, fixed positions of the satellite constellation, and labeled data restriction.

First, while our proposed method avoids sharing the raw data, the synthetic dataset generated by DP-CDM in PS is inevitably transferred to the local GS, satellite and edge router nodes. Although this transfer does not cause data leakage directly, it still contains the original functional features cut from the local data in PS. Intruders can backtrack these data to reproduce the original traffic recorded by PS. To make matters worse, they may infer the data from other nodes based on this traffic by exploring the distribution relationship. Basically, the data in PS is inevitably unsafe due to the exposure of the synthetic dataset. However, the data leakage in other nodes can be avoided, where one available mitigating solution is selecting the real samples with an evidently different data distribution compared with the local data in PS. These samples can be designated as public, making it challenging for intruders to infer other local data based on them. By adopting this approach, data leakage in nodes except PS is prevented by tackling the non-IID challenge in STIN intrusion detection.

Another limitation is that the dynamic topology nature of the STIN is not considered in STINIDF training. The satellite constellation in Figure 2 is indeed rotating around the earth in regular, each satellite is running on different designated orbits and possesses a dynamical connection condition with GSs. When the weight of DP-CDM in each local GS is uploaded to PS for overall aggregation through ISL-based satellites, its arriving order becomes extremely complicated and causes an asynchronous weight aggregation in PS. This asynchronous aggregation may make final aggregated DP-CDM unstable, lowering the final detection performance in the end.

Finally, the performance of STINIDF relies heavily on STI dataset, a labeled dataset collected in a designated simulation scenario. In real-world STIN scenarios, traffic collectors generally obtain massive unlabeled instances, unable to be applied in NIDS training. In this case, several unsupervised strategies like clustering and association are recommended to analyze data characteristics, behavior pattern, and feature distribution. Based on these information, a rule-based labeling mechanism can be designed to label the malicious and benign ones as the ground truth for the following supervised strategies.

VIII. Conclusion

In this paper, a satellite terrestrial integrated (STI) traffic dataset is collected to simulate the skewed data distribution in STIN, whereas a satellite terrestrial integrated network intrusion detection framework (STINIDF) is proposed to detect malicious traffic over this dataset. Based on federated learning with diffusion privacy, STINIDF first aggregates a conditional diffusion model that contains the global traffic information of STIN based on the local ground station and the edge router nodes. Then, it utilizes this model to aid the training process of the detection model in each node through data augmentation. The implemented experiments illustrate that STINIDF possesses 96.63% accuracy, 96.71% precision, 96.54% recall, and 96.66% F1 score in the designed non-IID STIN node scenario, which outperforms other state-of-the-art methods. With a privacy budget that equals 1, STINIDF can get 96.14% accuracy and generate instances with 17.88 FID, which also outperforms other synthetic data-assisted methods. In the future, two aspects are taken into consideration. First is the achievement of the dynamic topology for the satellite constellation, mentioned in the limitation section. Second is to enable NIDS with the capability on detecting L7-based DoS/DDoS. Compared with the ones based on the network layer, L7-based DoS/DDoS can falsify legitimate requests under the protocols like HTTP and HTTPS, causing a huger damage on target applications and servers with a small scale. Designing a detection strategy to prevent them is able to further improve the security of STIN.

Appendix

Algorithm 2 locally update DP-CDM in edge routers, local GSs, and PS with $E/E'/E''$ batch epochs. Based on Equation (9), line 6 defines a conditional loss function $\mathcal{L}$ given label y. Line 8 clips the gradient and line 10 adds the noise for (ϵ, δ)-DP utilization, corresponding to Equation (3).

Algorithm 2 DP-CDM Local Update Algorithm

Input: Local data distribution $p(x, y)$; DP-CDM z_w; batch size B; batch epoch $E/E'/E''$; time step N; Clip C; DP scale σ;

1: **for** $e = 1 : E/E'/E''$ **do**
2: $\{x, y\}_{b=1}^{B} \sim p(x, y)$
3: $\{x^{func}, x^{nonf}, y\}_{b=1}^{B} \leftarrow \{x, y\}_{b=1}^{B}$ // Functional features division
4: **for** every $\{x^{nonf,(b)}, y^{(b)}\}$ **do**
5: $n \sim \text{Uniform}(\{1, \ldots, N\})$, $z \sim \mathcal{N}(0, \mathrm{I})$
6: $\mathcal{L}_w^{(b)} = \|z - z_w(\sqrt{\bar{\alpha}_n} x^{nonf,(b)} + \sqrt{1 - \bar{\alpha}_n} z, n, y^{(b)})\|^2$
7: $g^{(b)} = d\mathcal{L}_w^{(b)}/dw$
8: $\overline{g}^{(b)} = g^{(b)} / \max(1, \|g^{(b)}\|_2 / C)$ // Clip gradient
9: **end for**
10: $\widetilde{g} = (\sum_B \overline{g}^{(b)} + \mathcal{N}(0, \sigma^2 C^2 \mathrm{I}))/B$ // Add noise
11: $w = w - \eta_w \widetilde{g}$ // Model update
12: **end for**

Algorithm 3 uses DP-CDM to generates traffic based on local data $p(x, y)$ in PS. In line 5 ~ 9, it reverses Gaussian vector x_N^{nonf} back to x_0^{nonf} for non-functional feature generation like a N-step DM-based backward process. Line 8 is based on Equation (8) while z_w requires label y for conditional generation.

Algorithm 3 DP-CDM Generation Algorithm

Input: Local data distribution $p(x, y)$; DP-CDM z_w; generation amount $|D'|$; time step N;

1: Create *newList*
2: **for** $m = 1 : |D'|$ **do**
3: $\{x, y\} \sim p(x, y)$
4: $\{x^{func}, x^{nonf}, y\} \leftarrow \{x, y\}$ // Functional features division
5: $x_N^{nonf} \sim \mathcal{N}(0, \mathrm{I})$
6: **for** $n = N : 1$ **do**
7: $z \sim \mathcal{N}(0, \mathrm{I})$ if $n > 1$, else $z = 0$
8: $x_{n-1}^{nonf} = \frac{1}{\sqrt{\alpha_n}}(x_n^{nonf} - \frac{1-\alpha_n}{\sqrt{1-\bar{\alpha}_n}} z_w(x_n^{nonf}, n, y)) + \beta_n z$
9: **end for**
10: $\bar{x} = x^{func} \oplus x_0^{nonf}, \bar{y} = y$
11: $newList.append(\bar{x}, \bar{y})$
12: **end for**
13: Return *newList*

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